AI FAQ ASSISTANT API

An AI-powered REST API for managing Frequently Asked Questions

Project Presentation

Technology: Node.js, Express.js, MongoDB, JWT & Google Gemini AI

INTRODUCTION

AI FAQ Assistant API is a backend application designed to manage FAQs through REST APIs.

It allows users to register and log in securely using JWT authentication.

Users can create, view, update, delete and search FAQ records.

Google Gemini AI is integrated to generate AI-based FAQ responses.

PROBLEM STATEMENT

Traditional FAQ systems often depend on static question-and-answer pages.

Finding the required information can be difficult when the FAQ collection becomes large.

Manual FAQ creation and management can be time-consuming.

A secure and intelligent API is needed to manage FAQs and provide AI-assisted answers.

EXISTING SYSTEM

Static FAQ pages or manually maintained FAQ databases.

Limited search and response capabilities.

No AI-based answer generation.

Manual management of questions and answers.

Security and user authentication may be limited.

PROPOSED SYSTEM

Develop a RESTful API for complete FAQ management.

Provide secure user registration and login using JWT.

Store users and FAQs in MongoDB.

Provide FAQ search functionality.

Integrate Google Gemini AI for intelligent FAQ generation and assistance.

OBJECTIVES

To develop a secure and scalable FAQ REST API.

To implement user authentication and authorization.

To perform CRUD operations on FAQ data.

To provide fast FAQ search functionality.

To integrate Generative AI for intelligent responses.

To maintain FAQ data using MongoDB.

TECHNOLOGIES USED

Node.js – JavaScript runtime environment

Express.js – REST API and server framework

MongoDB – NoSQL database

Mongoose – MongoDB object modeling

JWT – Authentication and authorization

bcrypt – Password hashing

Google Gemini AI – AI-powered FAQ generation

Postman / Thunder Client – API testing

SYSTEM ARCHITECTURE

Client / API Testing Tool

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Express.js REST API

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Authentication & Error Middleware

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Routes → Controllers / Services

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MongoDB Database

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Google Gemini AI

MODULES

1. User Authentication – Register, Login and JWT token generation
2. FAQ Management – Create, Read, Update and Delete FAQs
3. FAQ Search – Search FAQs using keywords
4. AI Assistant – Generate AI-based FAQ content/responses
5. Database Module – Store users and FAQ records
6. Error Handling – Manage validation and server errors

DATABASE DESIGN

User Collection:

- Name, Email, Password, Created Date

FAQ Collection:

- Question, Answer, Category, Created By, Created Date

MongoDB is used because it provides flexible document-based storage.

API FEATURES

POST /register – Create a new user account

POST /login – Authenticate user and generate JWT

GET /faqs – Retrieve FAQ records

POST /faqs – Create a new FAQ

PUT /faqs/:id – Update an FAQ

DELETE /faqs/:id – Delete an FAQ

GET /faqs/search – Search FAQs

AI route – Generate AI-assisted FAQ content

AI INTEGRATION

Google Gemini AI is connected with the backend API.

User input is sent to the AI service through a protected API route.

Gemini processes the prompt and generates a relevant response.

The generated response can be used for creating or assisting FAQ content.

This reduces manual effort and improves the intelligence of the FAQ system.

AUTHENTICATION & SECURITY

User passwords are protected using bcrypt hashing.

JWT tokens are generated after successful login.

Protected routes verify the JWT token before allowing access.

Environment variables are used for sensitive configuration values.

Middleware is used for authentication and centralized error handling.

TESTING

API endpoints are tested using Postman / Thunder Client.

Registration and login are tested with valid and invalid inputs.

JWT-protected routes are tested with authentication tokens.

FAQ CRUD operations are verified.

Search functionality is tested with different keywords.

AI response generation is tested using sample prompts.

ADVANTAGES

Easy and secure FAQ management.

RESTful API structure is simple to integrate with frontend applications.

JWT provides secure authentication.

MongoDB provides flexible data storage.

AI integration provides intelligent FAQ assistance.

Can be extended for web and mobile applications.

FUTURE ENHANCEMENT

Develop a web-based frontend for the API.

Add admin dashboard and role-based access control.

Add multilingual AI responses.

Improve semantic FAQ search using embeddings.

Add analytics and usage reports.

Deploy the API to a cloud platform for public access.

CONCLUSION

AI FAQ Assistant API provides a secure and intelligent backend solution for FAQ management.

It combines REST APIs, MongoDB, JWT authentication and Generative AI.

The system reduces manual FAQ management and provides AI-assisted responses.

The project can be further extended into a complete AI-powered customer support platform.

THANK YOU

Thank You

Questions & Answers